

TheraBank Loan Conversion Model

Technical Methodology, Data Processing, and Results Analysis

Executive Summary: This document outlines the methodology and results of a predictive classification model developed for TheraBank. The objective was to identify existing deposit customers with the highest propensity to convert to personal loan products. Leveraging **Weight of Evidence (WoE)** transformations and an interpretable **Logistic Regression** framework, the model achieved a highly robust validation accuracy of 96.8% and an AUC-ROC of 0.9690.

1. Data Processing & Methodology

To ensure high interpretability and handle non-linear relationships effectively, the data engineering pipeline utilized a rigorous **Weight of Evidence (WoE)** and **Information Value (IV)** framework prior to modeling.

Why WoE & IV?

- Handling Non-Linearity:** Continuous variables were binned (e.g., using quintiles), allowing the model to capture non-linear trends without complex polynomial transformations.
- Outlier & Null Robustness:** Blank values and extreme outliers are naturally handled by categorizing them into distinct bins and calculating their specific WoE.
- Feature Selection:** IV serves as a highly effective filter to eliminate noisy variables that offer no predictive power, reducing model complexity and preventing overfitting.

Feature Selection Results

Features with an IV score greater than 0.1 were deemed to have "medium to strong" predictive power and were selected for the final logistic regression model. Variables with $IV \leq 0.1$ were systematically dropped.

Feature	Information Value (IV)	Predictive Strength	Model Status
Income (in K/month)	4.2352	Extremely Strong	SELECTED
CCAvg	1.6072	Extremely Strong	SELECTED

CD Account	0.6152	Strong	SELECTED
Education	0.2992	Medium	SELECTED
Family members	0.0669	Weak	DROPPED
Mortgage	0.0668	Weak	DROPPED
Age / Experience / Securities / etc.	< 0.0100	Negligible	DROPPED

2. Model Architecture & Pipeline Walkthrough

The codebase implements a streamlined, production-ready pipeline utilizing Python (Pandas, Scikit-Learn):

- Data Cleansing:** Initial parsing to strip hidden whitespace in column names and removal of non-predictive identifiers (ID, ZIP Code). Handled negative spurious values.
- WoE Mapping Engine:** Iterates through all features. Continuous variables are sliced into quintiles. WoE is calculated as $\ln(\text{Distribution of Non-Events} / \text{Distribution of Events})$.
- Transformation:** The raw data space is mapped strictly into the WoE continuous space, linearizing the log-odds for the downstream model.
- Logistic Regression Training:** An L2-regularized Logistic Regression classifier is trained on an 80/20 stratified split.

3. Model Performance & Evaluation

The model was evaluated against an unseen holdout set (20%). The usage of WoE features yielded exceptional predictive separation.

0.9690

AUC-ROC

96.8%

ACCURACY

0.8106

KS STATISTIC

0.9380

GINI
COEFFICIENT

Confusion Matrix Analysis

The model is highly effective at minimizing False Positives while successfully identifying a large portion of the true loan converters.

	PREDICTED: NO LOAN	PREDICTED: LOAN
ACTUAL: NO LOAN	891	13
ACTUAL: LOAN	19	77

Classification Breakdown: The precision for identifying Loan converters is **86%** with a recall of **80%**. The model is highly reliable when it flags a customer as a potential prospect.

4. Interactive Prediction & Business Application

The script concludes with a dynamic prediction mechanism. It prompts the loan officer for the selected high-value inputs (Income, CCAvg, Education, CD Account), automatically transforms them via the generated WoE mapping dictionaries, and outputs a probability score.

Simulated Customer Scenario:

Inputs: Income: \$120K/mo, CCAvg: 2.0, Education: Higher (2), CD Account: Yes.

Result: The model outputs a **96.1% probability** of loan conversion. By restricting predictions only to highly informational variables ($IV > 0.1$), the business can target campaigns efficiently with less data collection friction.